Name: Meera Nair | USN / Roll No: 1PH23CS052

Subject: Engineering Physics - IA 1 | Section: C

1)

I law (inertia): no net external force means no change in the state of rest or uniform straight-line motion.

II law: $F = dp/dt = ma$.

III law: forces always occur in equal and opposite pairs acting on different bodies. Eg: a swimmer pushes water back and moves forward.

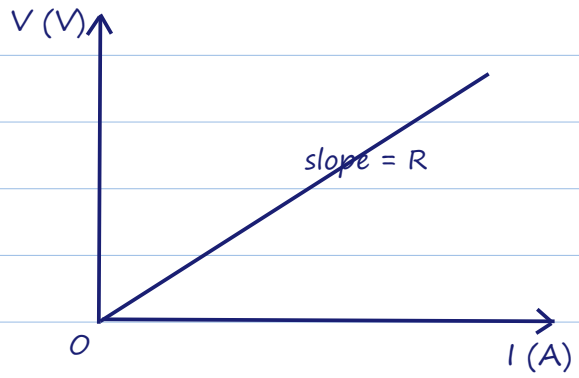
2)

(a) $a = (v - u)/t = 20/5 = 4\ m/s^2$

(b) $s = (u + v)/2 \times t = (0 + 20)/2 \times 5 = 50\ m$

3)

At constant temperature I is proportional to V, i.e. $V = IR$.



Straight line through origin; slope = $V/I = R$.

4)

KE = 1/2 mv^2 (due to motion), PE = mgh (due to position).

v^2 = u^2 + 2gh = 0 + 2 x 10 x 20 = 400, so v = 20 m/s.